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TATA POWER-DDL
Towards a Greener Tomorrow

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Digital Energy Grid (DEG): Built on Unified Energy Interface (UEI)

Presented By

Shashi Bala, DGM, PowerXchange Innovations Pvt Ltd

About PowerXchange

PowerXchange is an energy-technology company building **India's first Energy Super App**, designed to offer multiple digital energy programs that empower consumers with greater choice and control over how they produce, buy, and use electricity.

One of its flagship initiatives is a large-scale Peer-to-Peer (P2P) energy trading platform—a digital, consumer-centric energy marketplace powered by AI and blockchain technologies. Built on the Unified Energy Interface (UEI) architecture, the platform enables consumers and prosumers to directly buy and sell solar power in a transparent, secure, and near real-time settlement environment, while remainingfully aligned with existing DISCOM regulatory frameworks.

Experience of Powerxchange (Implementing Agency)

1. Powerxchange received the letter of award (LOA) from UPPCL for the implementation of P2P energy trading across discoms & Organisation deployed the solution.
2. Powerxchange was selected by REC Limited for Inter-state P2P Energy trading (As one of the first use-case of India Energy Stack) pilot between Tata Power-DDL, BSES & PVVNL.
3. Powerxchange platform was demonstrated in front of Honourable Prime Minister Shri Narendra Modi in Global Impact AI Summit New Delhi on 16th Feb 2026.
4. Powerxchange platform was demonstrated in front of Honourable Power Minister Shri Manohar Lal Khattar in Global Impact AI Summit New Delhi on 18th Feb 2026.

Peer-to-Peer (P2P) Trading of Rooftop Solar Energy – Pilots in India

ISGF conducted 3 successful pilot projects of P2P Trading of Rooftop Solar Energy on blockchain platform in India between 2020 and 2022

1. Uttar Pradesh (UP), India:

- Pilot implemented in 2020 under the **regulatory sandbox** approach to test the technical feasibility and customers willingness to participate in P2P trading
- **12 participants** (9 prosumers and 3 consumers) in Lucknow City participated in the pilot project
- **Project Go-Live: Dec 2020**
- Pilot project price discovery was around **INR 5.60/kWh (Win-Win for Prosumers and Consumers**
- **UP Electricity Regulatory Commission (UPERC) issued P2P Guidelines in April 2023**

2. Delhi, India:

- P2P trading of rooftop solar energy with Tata Power Delhi Distribution Ltd (TPDDL) in 2021
- **117 participants** in the pilot project – 2 MW rooftop solar involved
- **Project Go-Live: January 2021**
- Based on the project results, and recommendations to Delhi Electricity Regulatory Commission (DERC) **Peer to Peer Energy Transaction Guidelines, 2024 was issued by DERC in June 2024**

3. Kolkata, India

- Built a P2P trading platform on blockchain for CESC Kolkata in 2022 and run a 6-month pilot with 1002 C&I Customers (26 MW PV installations);
- **Project Go-Live: August 2022**
- Formulated a viable business model for CESC

Definitions

Sl. No.	Term	Description
1	Prosumers	Who generates the electricity from rooftop solar
2	Consumers	Who consumers the electricity from rooftop solar
3	BAP (Beckn Application Platform)	Used by consumers to access and transact on the energy network.
4	BPP (Beckn Provider Platform)	Used by prosumers to offer energy on the network.
5	Order Book	A ledger jointly maintained by BAPs and BPPs, recording initial energy transaction contracts.
6	DISCOM	Electricity Distribution Company serving the project area.
7	NPs (Network Participants)	Entities acting as BAPs/BPPs in the transaction layer.
8	Transaction Layer	Digital interface of NPs using Beckn Protocol to facilitate P2P energy transactions.
9	Data Trust Layer	Blockchain-based layer for storing P2P transaction data and fulfillment records from MDMS.
10	MDM (Meter Data Management)	A central system that collects, stores, and validates energy consumption data from smart meters, making it available for billing, forecasting, and analytics.
11	AMISP (Advanced Metering Infrastructure Service Provider)	An entity that installs, operates, and maintains smart meters and the communication system, ensuring reliable meter data transfer to DISCOMs and MDM.

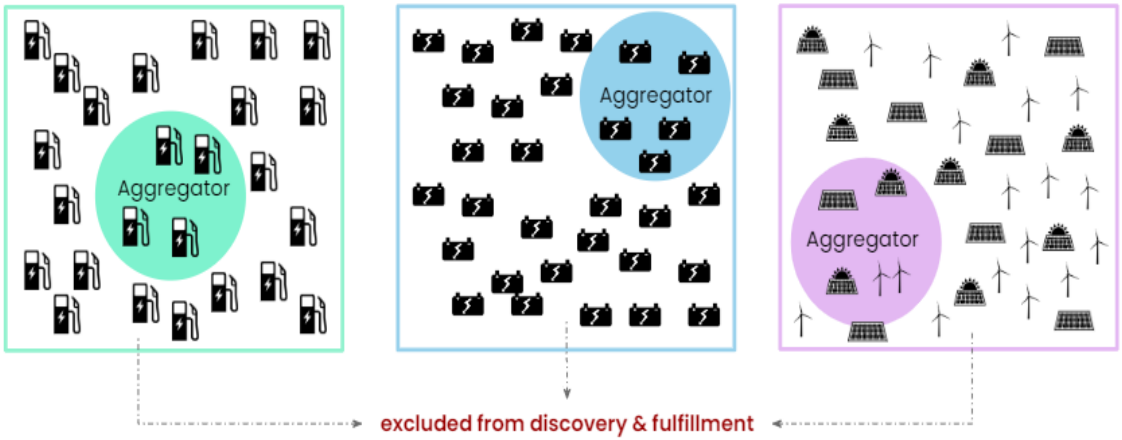
Unified Energy Interface (UEI) - Open Energy Networks

UNIFIED ENERGY INTERFACE

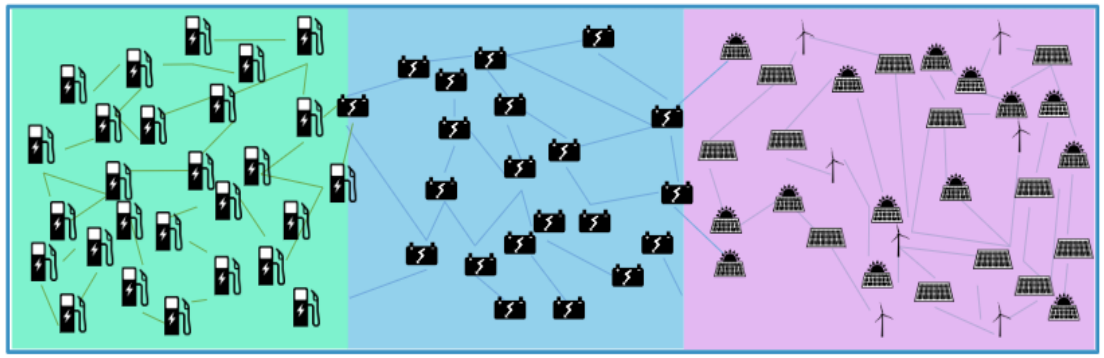
Unified Energy Interface (UEI) is an **open network for energy** to enable transactions between digital energy systems

Solves for

Discovery
Ordering
Fulfilment
Post-Fulfilment

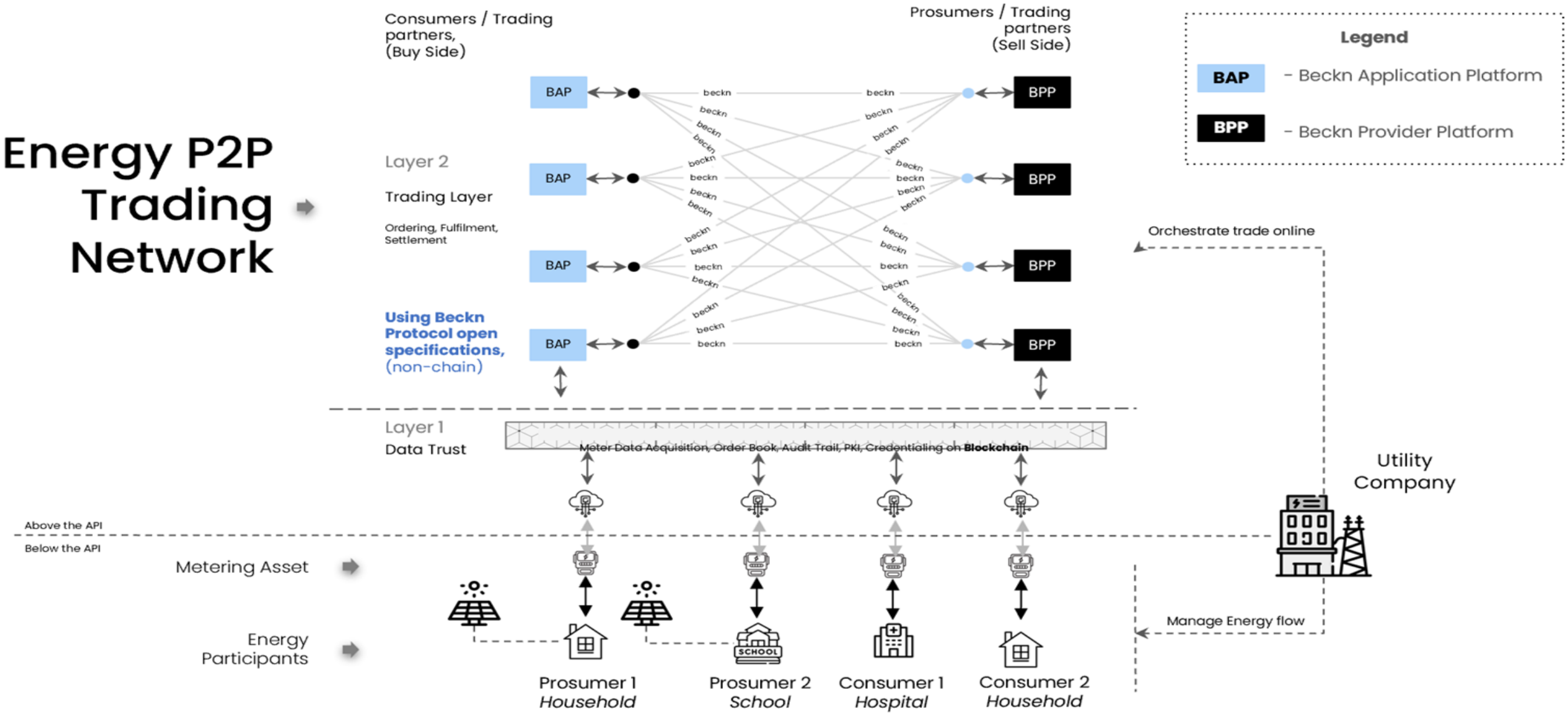


Move from Proprietary Platforms to Open Networks



Market Expansion | Autonomy at Nodes | Asynchronous & Combinatorial innovation

Energy P2P Trading Network on UEI Architecture



Methodology

1. Consumer Verification

Verification of consumer account details with DISCOM

Validation of smart meter and connection eligibility

2. P2P Trading Onboarding

Consumer registered as a P2P participant

DISCOM system updated for P2P trading

4. Meter Data Sharing

Smart meter readings shared with P2P platform via API/SFTP as preferred by the MDM System

Energy consumption and generation tracked

3. MDM Activation & User Marking

MDM system informed about P2P onboarding

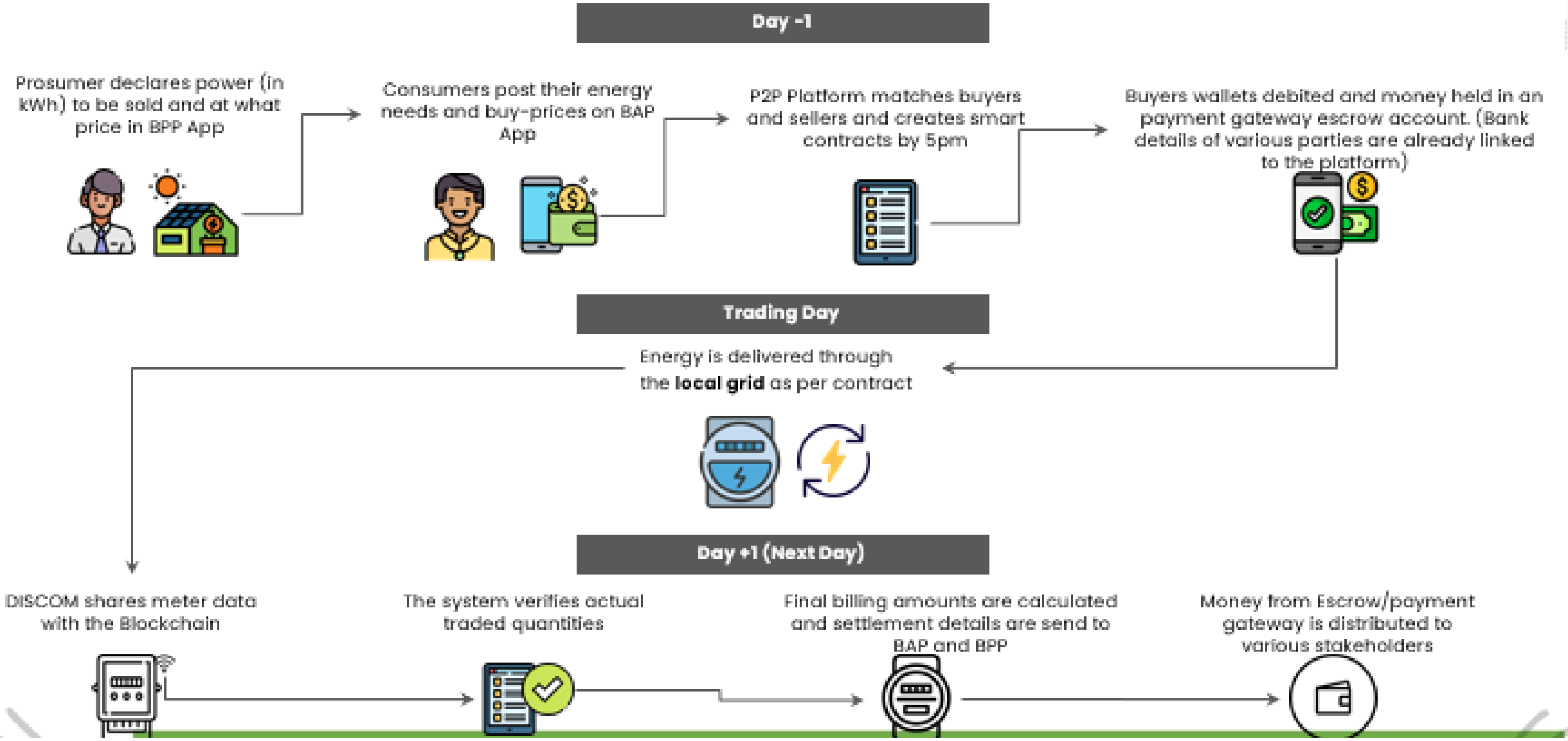
User marked in MDM to initiate meter data sharing

5. Settlement & Billing Synchronization

Traded energy units calculated for each user

Settlement shared with DISCOM for billing adjustment

Daily Trading (Day -1 to Day +1)



Monthly Billing and Reconciliation

Blockchain platform maintains all trade records of each P2P Participant

At month end (Month + 1 day), consolidated trade data is sent via SFTP from Blockchian to RMS for adjustments in the billing syatem

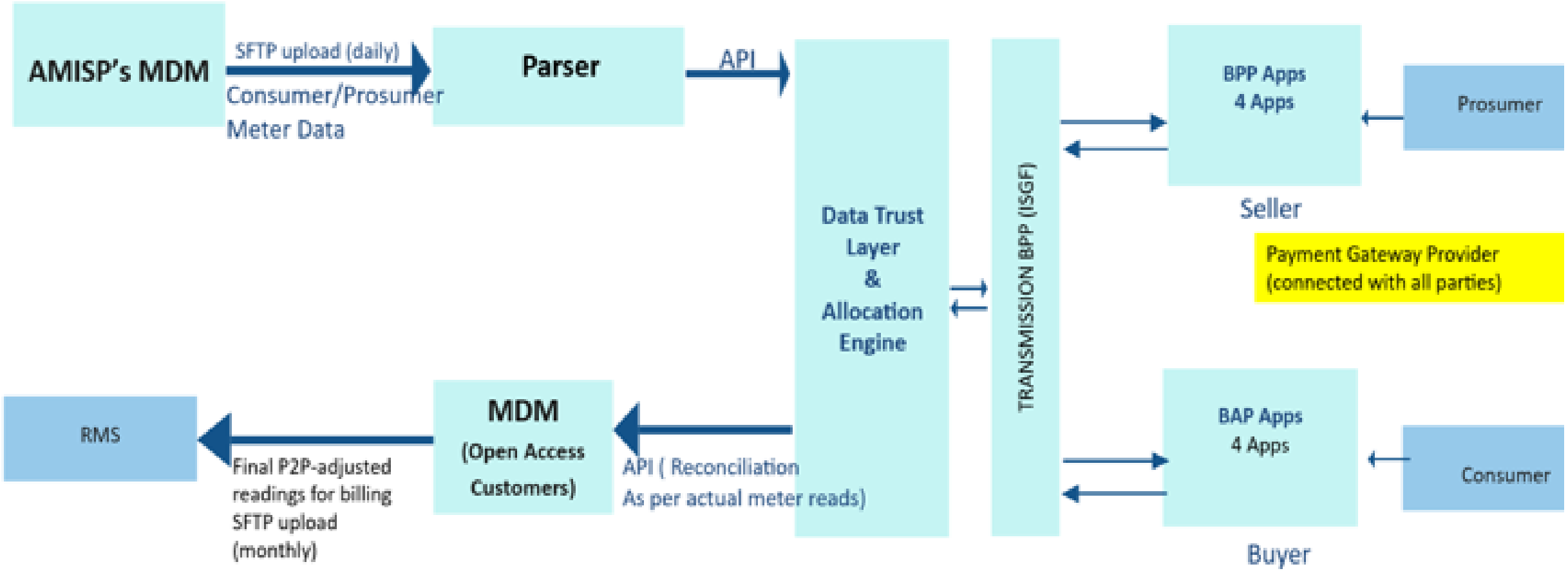
Adjustments are made in bills

Consumers: Energy bought on the P2P is deducted from total usage

Prosumers: Energy sold through P2P is deducted from solar energy exports

Discom monthly electricity bills generated accordingly

UEI System Architecture of UPPCL Demonstration Project



14 different systems interconnected and exchanging data successfully

Thank You